



HACETTEPE UNIVERSITY

**DEPARTMENT OF ELECTRICAL AND
ELECTRONICS ENGINEERING**

ELE489

Machine Learning

HW2

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Banknote Authentication using Decision Trees

This project implements a **Decision Tree Classifier** to authenticate banknotes based on image-derived statistical features. The dataset is sourced from the UCI Machine Learning Repository.

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Project Overview

The goal is to classify banknotes as **authentic (1)** or **fake (0)** using four numerical features extracted from wavelet-transformed images: 1. Variance 2. Skewness 3. Kurtosis 4. Entropy

We use a decision tree algorithm (`sklearn.tree.DecisionTreeClassifier`) to train and evaluate the model. Key steps include: - Exploratory data analysis (EDA) with pairwise feature visualization. - Hyperparameter tuning (`max_depth`, `criterion`, `min_samples_split`). - Model evaluation using accuracy, precision, recall, F1-score, and confusion matrices. - Visualization of the decision tree and feature importances.

Dataset

Source: UCI Banknote Authentication Dataset

Samples: 1,372 instances (762 authentic, 610 fake)

Features: 1. **variance:** Spread of pixel values. 2. **skewness:** Asymmetry of pixel distribution. 3. **kurtosis:** “Tailedness” of pixel distribution. 4. **entropy:** Complexity/randomness of the image.

Features Calculated

1. **Variance:** Measures how far pixel values are from the mean.
2. **Skewness:** Quantifies asymmetry in the pixel distribution.

- Positive skew: Tail on the right.
 - Negative skew: Tail on the left.
3. **Kurtosis**: Indicates heaviness of the tails in the distribution.
 4. **Entropy**: Measures randomness/complexity (higher entropy = more texture).
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Installation

1. **Google Colab**:

The notebook is designed to run directly in Google Colab. No local installation is required.

2. **Local Setup (Optional)**:

Clone the repository and install dependencies:

```
git clone https://github.com/your-username/banknote-authentication.git
cd banknote-authentication
```

Results

The decision tree model achieves high accuracy (~98%) in classifying banknotes as authentic or fake. Key metrics include precision, recall, and F1-scores exceeding 97% for both classes. The confusion matrix shows minimal misclassifications, with most predictions aligning with the true labels. Feature importance analysis reveals **variance** and **entropy** as the most critical predictors. A shallow tree (e.g., `max_depth=3`) balances interpretability and performance, showing clear decision rules.

Repository Structure

The repository contains:

- `decision_tree.ipynb`: The complete Colab notebook with code, visualizations, and analysis.
 - `README.md`: This documentation file.
 - `images/` (optional): Folder containing output plots like the decision tree visualization and confusion matrix.
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Acknowledgments

- Dataset: UCI Machine Learning Repository.
- Tools: Built with `scikit-learn`, `pandas`, `matplotlib`, and `seaborn`.